COMPUTER SYSTEMS & SECURITY ARCHITECTURE

From Bits to Breaches

Von Neumann Architecture, Abstraction Layers, the CIA Triad, and Threat Modeling

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PART 1

Von Neumann Architecture

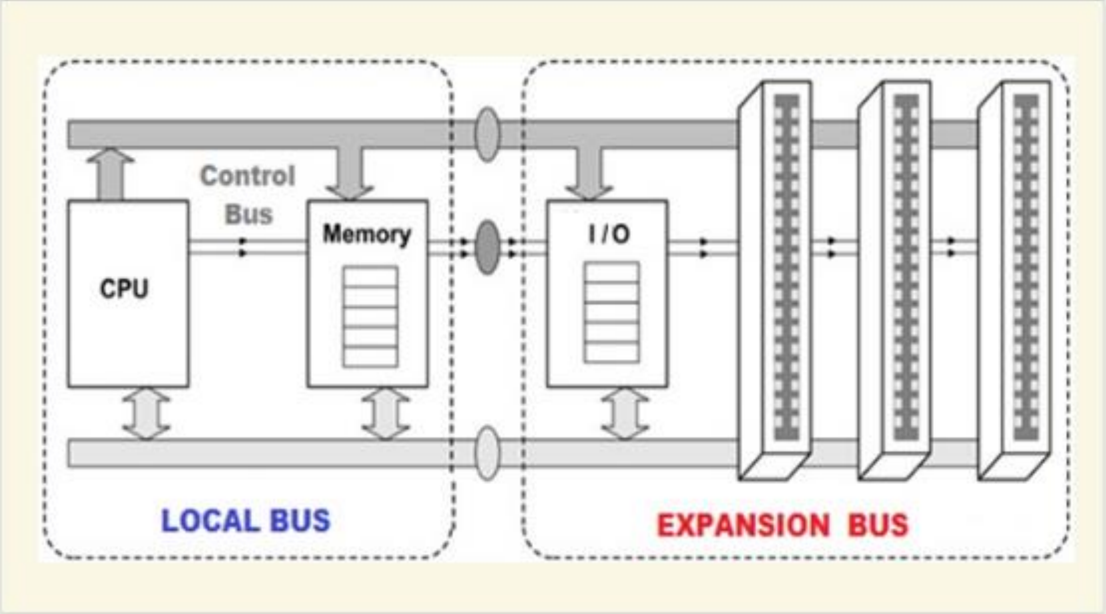
Understanding the Stored-Program Model, Hardware Cycles, Bottlenecks, and Fundamental Security Implications

Von Neumann Architecture

Core Hardware Components

- **Stored-Program Model:** A single read-write memory holds both executable instructions and operational data.
- **Central Processing Unit (CPU):** Houses the Control Unit (CU) and Arithmetic Logic Unit (ALU).
- **Fetch-Decode-Execute Cycle:** PC tracks instruction addresses; IR stores opcodes; ALU executes operations.
- **The Von Neumann Bottleneck:** Shared memory bus creates a throughput limit between CPU speed and memory latency.

System Bus Sections



Hardware Security & Mitigations

Architectural Contrast

Harvard Architecture: Physically separate memory and buses for instructions vs. data.

Modified Harvard: Modern CPUs use separate L1 instruction and data caches internally while exposing a unified address space to software.

Exploits & Hardware Mitigations

Code/Data Attacks: Buffer overflows inject executable shellcode into data regions due to unified memory.

Defenses: NX/DEP (No-Execute), W^X (Write XOR Execute), ASLR (Address Layout Randomization), and Stack Canaries.

PART 2

Computing Abstraction Layers

Managing System Complexity, Understanding Leaky Abstractions, and Mapping Attack Surfaces across the Stack

Computing Abstraction Layers

1. Hardware & Logic

Transistors, logic gates, and microarchitecture (pipelines, caches, branch predictors).

ISA Contract: x86-64 / ARMv8 interface visible to compilers.

2. Operating System

Kernel, virtual memory management (MMU), process scheduling, and system calls.

Runtimes: System libraries (libc), JVM, and language execution engines.

3. Application Tier

High-level languages (C++, Rust, Python), frameworks, and domain business logic.

Deployment: Hypervisors, containers, and cloud orchestrators.

PART 3

The CIA Triad

The Core Pillars of Information Assurance, Security Controls, and Operational Trade-off Decisions

The CIA Triad Breakdown

Confidentiality

Goal: Prevent unauthorized disclosure of information.

Key Controls: AES Encryption (at rest & transit), RBAC/ABAC access controls, tokenization, and strict secrets management.

Integrity

Goal: Prevent unauthorized modification; guarantee accuracy.

Key Controls: Cryptographic hashes (SHA-256), digital signatures, code signing, and immutable audit logs.

Availability

Goal: Ensure timely and reliable access to systems & data.

Key Controls: High-availability failover, autoscaling, rate-limiting, DDoS mitigation, and tested backups.

PART 4

Threat Modeling

Structured Methodologies for Systematically Identifying, Prioritizing, and Mitigating Software Risks

Threat Modeling Process

1

Scope & Assets

Define high-value data, user roles, and potential adversaries.

2

System DFDs

Map Data Flow Diagrams and identify trust boundaries.

3

STRIDE Analysis

Categorize threats against each architectural element.

4

Risk & Mitigate

Calculate Risk = Likelihood × Impact; implement controls.

STRIDE Threat Framework

Category	Target Property	Example Attack Threat	Primary Countermeasure
Spoofing	Authenticity	Credential stuffing, session hijacking	MFA, secure cookie flags, OAuth 2.0
Tampering	Integrity	SQL injection, parameter tampering	Parameterized queries, input validation
Repudiation	Non-repudiation	User denies critical financial transfer	Tamper-evident, time-stamped logging
Info Disclosure	Confidentiality	S3 bucket leakage, TLS misconfig	Least-privilege IAM, TLS 1.3 encryption
Denial of Service	Availability	Layer 7 flood, expensive SQL query	Rate limiting, caching, DDoS protection
Elevation of Privilege	Authorization	Buffer overflow, SSRF metadata access	Safe deserialization, IMDSv2, RBAC

Questions & Discussion

From Bits to Breaches: Building Resilient Systems Through Layered Defense

📘 Computer Science & DevSecOps Series Systems Architecture Module 🎓

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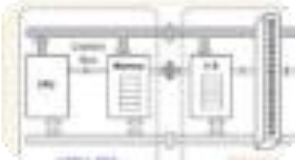

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